Removing the Weyl hypothesis by the topology of ALE embeddings

24 September 2026. This note supplies the previously missing input in the compact-regime realization proof. The stronger ALE nonembedding theorem is **an existing result**, not a new classification discovered here: it is Zhang's theorem with the noncompact extension stated explicitly by Bamler–Chen. Its application changes the scope of the project's realization theorem substantially.

In particular, the residual possibility recorded in the preceding topology note, namely a mixed-chirality rational-ball bubble in the universal cover, can now be excluded. No classification of arbitrary Ricci-flat ALE metrics is assumed.

1. The classical input that closes the remaining bubble alternative

ALE nonembedding lemma (Zhang; Bamler–Chen). Let N be a smooth four-manifold, possibly noncompact, with

$$H_2(N;\mathbb{Z}) = 0, \qquad H_1(N;\mathbb{Z}) \text{ torsion-free.} \tag{1}$$

No compact smooth domain $B \subset N$ has interior diffeomorphic to a nonflat complete Ricci-flat ALE four-manifold.

The precise noncompact formulation is Bamler–Chen, Lemma 4.35, p. 41, which cites Zhang, Theorem 6.1 and Lemmas 6.2–6.5, pp. 378–382. In the published version of Bamler–Chen, the statement is still Lemma 4.35: DOI 10.1002/cpa.70024. The isolated $H_1(M)$ in its printed statement denotes the ambient $H_1(N)$, as its following proof makes explicit. We use the lemma only when this group is zero.

Here is the mechanism, to distinguish this input from the unavailable classification of all Ricci-flat ALE spaces. In an ambient with $H_1 = H_2 = 0$, write $Y = \partial B = S^3/\Gamma$ and $C = N \setminus \operatorname{int} B$. Integral Mayer–Vietoris gives

$$H_2(B;\mathbb{Z}) = 0, \qquad H_1(Y;\mathbb{Z}) \simeq H_1(B;\mathbb{Z}) \oplus H_1(C;\mathbb{Z}). \tag{2}$$

The ALE fundamental group is finite, and the end group surjects onto it. Thus $H_1(B,Y;\mathbb{Z}) = 0$. Duality and the universal coefficient theorem identify the kernel of $H_1(Y) \to H_1(B)$ with a group isomorphic to $H_1(B)$. The splitting in (2) respects this projection, so

$$H_1(Y) \simeq G \oplus G. \tag{3}$$

This is stronger than merely requiring that $|H_1(Y)|$ be a square.

Zhang's spherical-space-form calculation reduces the nontrivial possibilities in (3) to binary dihedral groups of order $8n$, $n \geqslant 1$, and the binary icosahedral group. His representation lemma and eta calculation give, up to orientation,

$$|\eta(Y)| = \frac{8n^2+1}{12n} \quad \text{or} \quad \frac{361}{180}, \qquad \text{respectively.} \tag{4}$$

Both exceed 2/3. On the other hand, $b_2(B) = 0$, finite first homology, connected boundary and duality give $\chi(B) = 1$, $\tau(B) = 0$. Ricci-flat Chern–Gauss–Bonnet and APS signature, in curvature-operator norms, imply

$$\begin{aligned} E_+ + E_- &= 8\pi^2\left(1 - |\Gamma|^{-1}\right), \\ E_+ - E_- &= 12\pi^2\eta(Y), \qquad E_\pm = \int |W^\pm|^2 \geqslant 0. \end{aligned} \tag{5}$$

Consequently

$$|\eta(Y)| \leqslant \frac{2}{3}\left(1 - |\Gamma|^{-1}\right) < \frac{2}{3}, \tag{6}$$

contradicting (4). The trivial end group gives zero total curvature in (5), hence flatness. This explains why a mixed-chirality rational ball does not evade the argument: nonnegativity of **both** energies is already incompatible with its boundary data. The argument does not set either Weyl half to zero.

For the quaternion test $\Gamma = Q_8$, (3) is indeed possible: the boundary has $H_1 = (\mathbb{Z}/2)^2$. It is (6), not this group calculation alone, that excludes a Ricci-flat rational-ball filling: $3/4 > 7/12$. This separates the topological and geometric restrictions.

2. Vanishing second homotopy suffices; torsion in the first is allowed

Theorem 1 (all-ALE-core exclusion). Let M be a connected smooth oriented four-manifold with $\pi_2(M) = 0$. No large compact core of a nonflat complete smooth Ricci-flat ALE four-manifold embeds smoothly in M. No torsion-free assumption on $\pi_1(M)$ is needed.

Proof. Let $i : B \hookrightarrow M$ be such an embedding and let $p : \widetilde{M} \to M$ be the universal cover. Hurewicz gives

$$H_1(\widetilde{M}; \mathbb{Z}) = H_2(\widetilde{M}; \mathbb{Z}) = 0. \tag{7}$$

The fundamental group of a Ricci-flat ALE space is finite. A component $\widehat{B}$ of $p^{-1}(i(B))$ is therefore a **finite** cover of B: its degree is the order of the finite image of $i_* : \pi_1(B) \to \pi_1(M)$. In particular, it is a compact smooth domain in $\widetilde{M}$. This step neither assumes that the core fundamental group injects nor that its image is trivial.

The interior of $\widehat{B}$ carries the lifted complete Ricci-flat metric. A finite cover of an ALE space has finitely many lifted ALE ends and Euclidean volume growth. It has only one end: if it had two, the nonnegative-Ricci splitting theorem would split it as a line times a compact manifold, contradicting Euclidean four-dimensional volume growth. Thus the lifted space is itself nonflat Ricci-flat ALE. Apply the classical lemma to its compact domain $\widehat{B} \subset \widetilde{M}$, using (7), to obtain a contradiction. □

For completeness, finiteness of the ALE fundamental group can also be seen directly from the end: take the cover corresponding to the image of $\pi_1(Y) \to \pi_1(B)$. This subgroup is finite. If its index were greater than one, one lifted end has degree one and is separated from the rest by a compact cross-section. The complete lifted metric would have at least two ends and Euclidean lower volume growth, again contradicting splitting. Therefore the end group surjects onto the full fundamental group. This reasoning uses Ricci-flatness and the ALE end, not any holonomy classification. The same end-cover/splitting argument appears in Wright, Section 3 and Proposition 1; only that argument is used, not the paper's half-flat classification.

The conclusion is strictly broader in topology than the previous condition π_1 torsion-free, $\pi_2 = 0$. For example, $L(p, q) \times S^1$ has vanishing π_2 and fundamental group $(\mathbb{Z}/p) \times \mathbb{Z}$; homotopy four-spheres also have vanishing π_2. These illustrate the topological scope, not a claim that every example has the near-Einstein inputs of the next theorem.

3. Bounded-geometry realization without gauge saturation or Weyl bounds

Theorem 2 (harmonic compactness). Fix $K, D, v > 0$ and $0 < \alpha < 1$. Closed connected smooth oriented four-manifolds with

$$\pi_2(M) = 0, \qquad \|\mathrm{Ric}^\sharp\|_{\infty,\mathrm{op}} \leqslant K, \qquad \mathrm{diam}(g) \leqslant D, \qquad V_g \geqslant v \tag{8}$$

have a uniformly positive $C^{1,\alpha}$ harmonic radius. They are precompact modulo diffeomorphisms in $C^{1,\beta}$, $\beta < 1$, and weak $W^{2,p}$, $p < \infty$, and have finitely many smooth types.

Proof. The analytic argument of compact realization, Sections 3–4 is used up to, but not including, its one-half Weyl step. Bishop–Gromov gives uniform noncollapse on fixed scales and a total-volume upper bound. Jiang–Naber supplies the full curvature-energy bound. If the minimum harmonic radius ρ_j tends to zero, rescale by ρ_j^{-2}. The normalized harmonic atlases converge, Ricci tends uniformly to zero, and the harmonic-coordinate metric equation upgrades convergence locally to strong $W^{2,p}$ for finite p. The limit is smooth Ricci-flat. Recovery of harmonic charts with strict margins excludes a flat limit.

Euclidean volume growth and finite curvature energy make this limit ALE by Bando–Kasue–Nakajima. Only positive-order ALE asymptotics are needed; no optimal decay-order assertion is used. Its smooth compact core embeds in each sufficiently late M_j, contradicting Theorem 1. There is **no need to force W^+ or W^- to vanish**. The remaining finite-atlas compactness argument is unchanged. □

Theorem 3 (same-smooth-type realization). Fix $D, v > 0$. There is $\varepsilon(D, v) > 0$, uniform over smooth types and $\sigma \in \{-1, +1\}$, such that

$$\begin{aligned} &\pi_2(M) = 0, \qquad \mathrm{diam}(g) \leqslant D, \qquad V_g \geqslant v, \\ &\delta_\sigma(g) := \|\mathrm{Ric}_g^\sharp - \sigma I\|_{\infty,\mathrm{op}} < \varepsilon(D, v) \end{aligned} \quad \Longrightarrow \quad \exists h \text{ on } M : \ \mathrm{Ric}_h = \sigma h. \tag{9}$$

Proof. Otherwise take counterexamples with the fixed data and $\delta_{\sigma_j}(g_j) \to 0$. Pass to a subsequence with constant sign. Theorem 2 applies with $K = 2$ to late terms and identifies them with a fixed smooth manifold. In harmonic coordinates the limiting metric satisfies $\mathrm{Ric} = \sigma g$ weakly. Elliptic regularity makes it smooth. Pull it back to the original smooth manifolds, contradicting their defining absence of such an Einstein metric. □

In the positive case, $\delta_+ < 1/2$ gives $\mathrm{Ric} \geqslant g/2$; Bonnet–Myers supplies $\mathrm{diam} \leqslant \pi\sqrt{6}$. Thus for each $v > 0$, $\pi_2 = 0$, $V_g \geqslant v$, and sufficiently small δ_+ already imply a positive Einstein metric, with threshold depending only on v. This still retains the noncollapse input.

On the negative side an actual conformally solvable monopole class with $(a_g^+)^2 \geqslant A_0 > 0$ gives

$$V_g \geqslant \frac{2\pi^2 A_0}{(1 + \delta_-(g))^2} \geqslant \frac{8\pi^2 A_0}{9} \quad (\delta_- \leqslant 1/2). \tag{10}$$

Hence (9) yields a threshold $\varepsilon(D, A_0)$ from this actual class input and diameter alone. Unlike the preceding KE realization theorem, **no smallness of $\mathfrak{D}_2/V$ is required**. The conclusion is negative Einstein, not necessarily Kähler–Einstein. That already satisfies the existence output of the adopted Yau26 problem.

4. What is resolved, and what is not

This closes the entire compact, noncollapsed bubble alternative on $\pi_2 = 0$ topology. A hypothetical negative-side Yau26 counterexample sequence in that class, with uniform positive actual projection budget, must have diameters tending to infinity: any bounded-diameter subsequence would contradict (9)–(10). No near-saturation is necessary for this escape conclusion.

On the positive side, if $\pi_2(M_j) = 0$, no M_j admits any Einstein metric, and $\delta_+(g_j) \to 0$, then necessarily $V_{g_j} \to 0$. Otherwise a subsequence has $V_{g_j} \geqslant v > 0$, and Myers plus Theorem 3 produces a positive Einstein metric on every sufficiently late member of that subsequence. Thus, within the $\pi_2 = 0$ topology, the two signs have different remaining escape mechanisms: positive candidates must collapse in total volume; negative candidates with a positive actual projection budget must escape in diameter.

The original dimension-only problem still allows diameter escape, collapse without a supplied volume budget, and topology with nonzero π_2. Those are not excluded here. The theorem does not decide whether the fixed non-Kähler shear examples admit general Einstein metrics: their known approximating sequence has diverging diameter.

The old near-saturation theorem retains its independent stronger geometric output: its actual forms recover a global parallel form and hence KE. For **general Einstein existence** in the compact regime, however, saturation and the one-sided Weyl assumptions are now unnecessary.

The mathematical increment in this note is the explicit composition of the full ALE nonembedding input, the finite-cover lift, and the preserved almost-Einstein compactness argument. Zhang's boundary/eta obstruction and Bamler–Chen's noncompact formulation are credited as existing results. No assertion of world-first priority for the resulting compact-regime corollary, or classification of all Ricci-flat ALE spaces, is made.

For the historical comparison, Anderson's survey, Corollary 5.8 already gives exclusion of the orbifold boundary of the exact Einstein moduli space under vanishing $H_2(M; \mathbf{F})$ for every field. Bamler–Chen explicitly credit Anderson's earlier work for the nonembedding idea. The composition here uses the later explicit noncompact lemma, the finite-cover lift under $\pi_2 = 0$, and bounded-Ricci harmonic compactness to handle varying smooth types and pointwise approximate Einstein inputs. This distinguishes the statements being used; it does not establish that this composed corollary was absent from the literature.